

Hostel Maintenance & Issue Ticketing System

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001 (GIVEN IN QUESTION)	Functional	The system shall enable residents to log maintenance tickets with room number tags, photo attachments, and categorized issue severity.	High	Pass: Ticket is created, assigned a unique ID, and routed to warden queue. Fail: Incomplete form accepted without room tag.	Establishes the entry point for all maintenance requests with enough context to triage correctly.	(given)
FR-002	Functional	Warden assigns each new ticket to a staff member who's free and handles that category — plumbing, electrical, or carpentry — instead of picking randomly.	High	Pass: status moves to "Assigned" and the staff member is notified within 2 min. Fail: ticket stays "Unassigned" after warden confirms it.	Saves the warden from manually tracking who's free and who's already swamped.	
FR-003	Functional	A ticket can only move forward through Logged → Assigned → In Progress → Resolved → Closed, in that order, and only the warden or assigned staff can push it along.	High	Pass: each change is timestamped, invalid jumps blocked. Fail: a ticket skips a stage, e.g. Logged straight to Closed.	Without this you can't trust the SLA numbers — someone could just mark things Closed to hide a backlog.	
FR-004	Functional	Resident has to confirm the fix before a ticket is really closed — they can also reject it if the issue's still there.	Medium	Pass: reaches "Closed" only after resident confirms; rejection reopens and sends it back to warden. Fail: ticket auto-closes with no resident input.	A resolved status from staff doesn't always mean the problem's actually gone.	
FR-005	Functional	Residents can pull up live status and full history for anything they've raised.	Medium	Pass: status shown matches backend within 5 sec of an update. Fail: mismatch between what resident sees and actual record.	Stops people from raising the same ticket twice just because they can't see it's already being worked on.	
NFR-001 (GIVEN UNDER QUESTION)	Non-Functional	High-priority ticket past 24 hrs with no status change triggers an automatic SMS/email escalation.	High	Pass: load testing confirms alert fires within target latency, holds up at peak traffic.	An electrical issue that just sits there silently is a safety problem, not just an inconvenience.	(given)
NFR-002	Non-Functional	Role-based access — residents only see their own tickets, wardens only see tickets from their own block.	High	Pass: cross-resident / cross-block access attempts blocked (HTTP 403). Fail: any unauthorized read/write goes through.	Room numbers and photos are personal — no reason a warden from another block should see them.	